De La Salle Holy Cross College

NOVEMBER EXAMINATION

2024

Grade 10

LIFE SCIENCES

Examiner:	Mr B Mlangeni	**Time:**	2.5 hours
Moderator:	Ms B Hannweg	**Marks:**	150

Name: -

_____**MEMO**_____

PLEASE READ THROUGH THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of 23 pages. Please check that your question paper is complete.
2. Answer ALL the questions on this question paper in the spaces provided.
3. It is in your interest to write legibly and to present your work neatly.

Questions	**1**	**2**	**3**	**4**	**5**	**Total**
Possible mark	65	28	19	24	14	150
Actual mark						

Remember that God is with you and will help and inspire you in all that you do

Section A

Question 1

1.1 Answer the following multiple choices questions by choosing the most correct answers from option A - D. Write the letter of the correct answer in the table below. (20)

Question number	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
Corresponding letter	C	B	B	A	B	C	C	D	B	C

1.2.1 Which process converts glucose into energy in the presence of oxygen?

a) Fermentation

b) Photosynthesis

c) Respiration

1) Mitosis

1.1.2 In which phase of mitosis do chromosomes align at the cell's equator?

a) Prophase

b) Metaphase

c) Anaphase

1) Telophase

1.1.3 Which nutrient is primarily responsible for providing energy to living organisms?

a) Proteins

b) Carbohydrates

c) Vitamins

1) Minerals

1.1.4 The process by which plants lose water vapor through small openings in their leaves is called:

a) Transpiration

b) Respiration

1) Evaporation

1) Condensation

1.1.5 The ecological term for the role an organism plays in its environment is called:

a) Habitat

b) Niche

c) Population

d) Community

1.1.6 The primary site of nutrient absorption in the human digestive system is:

a) Stomach

b) Large intestine

c) Small intestine

d) Liver

1.1.7 Which of the following statements about chloroplasts and mitochondria is correct?

a) Chloroplasts are responsible for aerobic respiration, while mitochondria carry out photosynthesis.

b) Both chloroplasts and mitochondria generate water but through different processes.

c) Mitochondria produce carbon dioxide as a byproduct, while chloroplasts produce oxygen.

d) Chloroplasts and mitochondria both function in the breakdown of glucose.

1.1.8 Which of the following statements about enzymes is true?

a) Enzymes are consumed during chemical reactions.

b) Enzymes increase the activation energy of a reaction

c) Enzymes can work under any temperature and pH.

d) Enzymes are highly specific and act on one type of substrate.

1.1.9 The fluid mosaic model describes:

a) The structure of DNA and RNA molecules.

b) The arrangement of proteins and lipids in the plasma membrane.

c) The formation of ATP in the mitochondria.

d) The process of transcription in the nucleus.

1.1.10 In the scientific method, which of the following describes the dependent variable?

a) The variable that the scientist changes.

b) The variable that remains constant throughout the experiment.

c) The variable that is measured in response to changes.

d) The variable that is used as a control.

1.2 Give the correct biological term for the following descriptions. (10)

1.2.1 The maintenance of a stable internal environment in living organisms.

_____

- Homeostasis

1.2.2 The movement of substances across a cell membrane against a concentration gradient, requiring energy.

_____

- Active transport

1.2.3 An organism capable of producing its own food through processes like photosynthesis.

_____

- Autotrophs/ Producers

1.2.4 The first stage of cellular respiration, where glucose is broken down in the cytoplasm.

_____

- Glycolysis

1.2.5 Small, finger-like projections in the small intestine that increase surface area for absorption of nutrients.

_____

- Microvilli

1.2.6 Organisms found in an ecosystem that cannot make their own food.

_____

- Heterotrophs

1.2.7 Condition where the concentration of solutes outside the cell is equal to the concentration of solutes inside the cell.

_____

- Isotonic/ equilibrium

1.2.8 The process by which a cell takes in big material by engulfing them to form an internal vesicle.

_____

- Endocytosis

1.2.9 Peptide hormone produced by the lining of the duodenum.

_____

- Secretin

1.2.10 A pyramid that represents the amount of energy available at each trophic level.

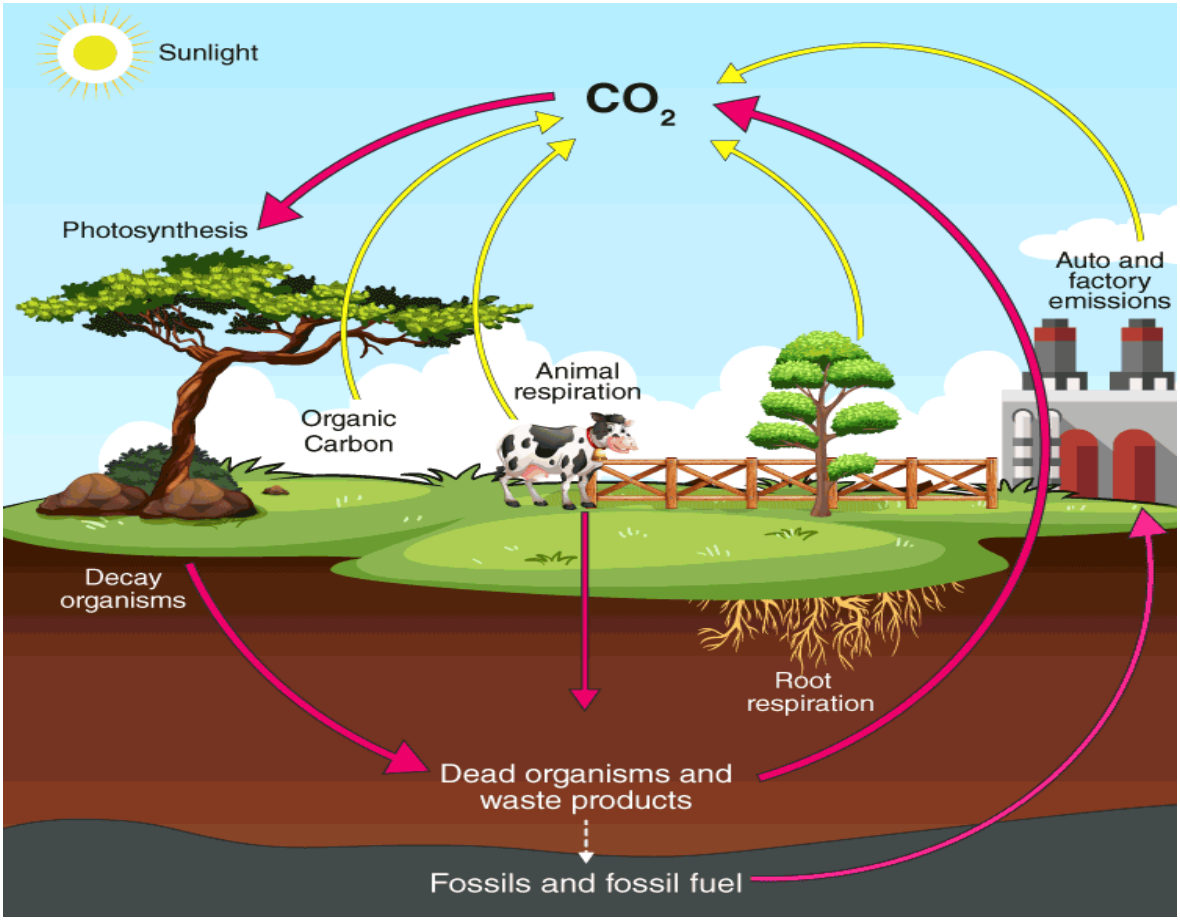
_____

- Energy pyramid

1.3 Select the term in Column A that best matches the description in Column B. Write the letter of the description in the Answer column provided. Each letter may be used only once. (10)

Column A	Answer	Column B
1. Hypothesis	G	A. Low blood sugar levels
2. Large intestine	K	B. Gel-like substance that is found inside the nucleus
3. Hypoglycemia	A	C. A single layer of plant cells containing few or no chloroplasts
4. Granum	H	D. Rhythmic, wavelike motion of cilia that propels particles through the airways
5. Prokaryotes	I	E. Stimulates bile production in the liver
6. Food chain	J	F. Liquid found inside the chloroplast
7. Peristalsis	L	G. A prediction a scientist makes about a phenomenon
8. Nucleoplasm	B	H. Stack of thylakoids
9. Stroma	F	I. Cells with no true nuclei
10. Epidermis	C	J. Shows the direction of flow of energy in an ecosystem
		K. Part of the gastro intestinal tract that absorbs water
		L. Wave-like contractions that start in the oesophagus

1.4 Use the information in Figure 1 on the carbon cycle to answer the questions.



1.4.1. Use the information in Figure 1 in addition to your reasoning, and show whether carbon is either release or removed from the atmosphere, by writing released or removed where appropriate in Table 1. (4)

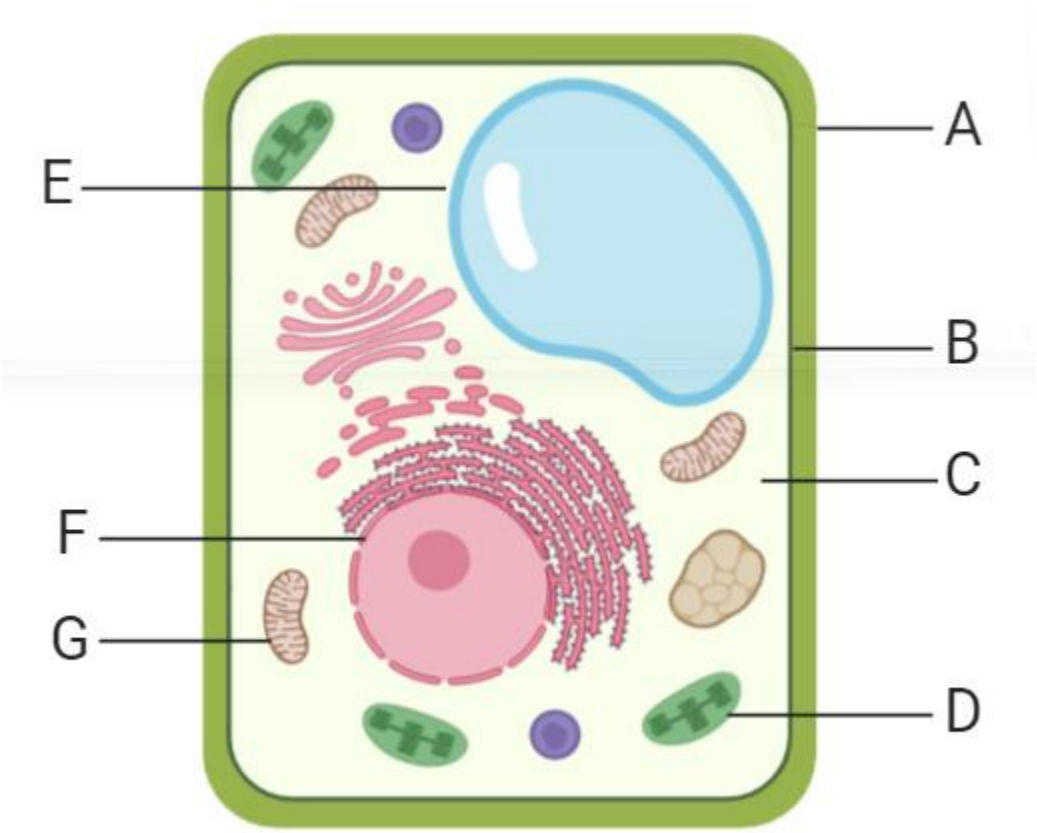
Table 1. Table showing the processes involved in the carbon cycle

Processes	**Carbon released/ removed from the atmosphere**
Decomposition	Released
Photosynthesis	Removed
Burning of fossil fuels	Released
Respiration	Released

1.4.2 Name the reactants and products of photosynthesis. (4)

- Reactants: Water + CO2/ sunlight
- Products: Glucose + O2

1.5 Study the diagram and answer the questions that follow.



1.5.1 Is this a plant or an animal cell? Give two visible reasons for your answer. (3)

- Plant cell
- Has chloroplast
- Has cell wall
- Has a regular shape
- Large vacoule
- Any two reasons

1.5.2 Label parts G, C and F on the diagram. (3)

- G- Mitochondria
- C- Cytoplasm
- F- Nucleus/ nuclear membrane/pore

1.5.3 Give the letter and name of the organelle that allows the cell to communicate with its environment and other cells. (2)

- B; cell membrane; also accept
- F- Nucleus

1.5.4 Give the name of the pigment found in part D. (1)

- Chlorophyll

1.5.5 Explain what makes part B selectively permeable. (3)

- Phospholipid bilayer/ hydrophilic and hydrophobic
- Allows small and uncharged particles to pass through it
- Does not allow big and charged particles to pass through it
- Protein pumps
- Diffusion
- Any reason that explains how particles move across the membrane

1.5.3 Study the following table that consists of two items (numbered 1 and 2) in the first column and a term in the second column. Decide which item(s) relate to the term. (5)

Write down your choice in the space provided in the ANSWER column, making use of the following codes:
A only item 1 relates to the term
B only item 2 relates to the term
C both items 1 and 2 relate to the term
D neither item 1 nor item 2 relates to the term

Items	Term	Answer
1. Essential in the process of photosynthesis 2. Only found in plants	Plastids	C
1. Can break down carbohydrates and lipids 2. Contains enzymes	Lysosomes	C
1. Fully permeable 2. Components are constantly moving past one another	Cell membrane	B
1. Uses energy and proteins to transport particles 2. Only occurs in animal cells	Facilitated diffusion	D
1. Adjacent to the nucleus 2. Is the site of protein synthesis	Smooth endoplasmic reticulum	D

[65]

Section B

Question 2

2.1 Read the following and answer the following questions

Lactose intolerance
April 7, 2023
Reviewed by Howard E. LeWine, MD, Chief Medical Editor, Harvard Health Publishing

What is it?

Lactose intolerance is a common cause of abdominal cramping, bloating and loose stools. This condition occurs when the body does not have enough of the intestinal enzyme lactase. The job of lactase is to break down lactose, the main sugar in milk. Once lactose is broken down into simpler forms of sugar, these simple sugars can be absorbed into the bloodstream.

Lactose intolerance happens when your small intestine does not make enough of a digestive enzyme called lactase. Lactase breaks down the lactose in food so your body can absorb it. In normal digestion, lactose is digested in the small intestine without the release of gas bubbles. When lactose can't be digested well, it passes into the colon. Bacteria in the colon break down some of the lactose, producing hydrogen gas. The remaining lactose also draws water into the colon. The extra gas and water result in symptoms, such as cramping, diarrhoea, bloating and flatulence (releasing gas).

Lactose intolerance usually is genetic (inherited). In many people of African or Asian descent, the body begins making less lactase around age 5. As many as 90% of people from some areas of Eastern Asia, 80% of American Indians, 65% of Africans and African-Americans, and 50% of Hispanics have some degree of lactose intolerance. In contrast, most Caucasians (80%) have a gene that preserves the ability to produce lactase into adulthood.

https://www.health.harvard.edu/a_to_z/lactose-intolerance-a-to-z

2.1.1 Lactose is a disaccharide sugar in milk. Name the monomers that make up lactose. (2)

_____

- Galactose
- Glucose

2.2 One of the dietary modifications for lactose intolerance is drinking small amounts of milk.

2.2.1 Explain how a lactose intolerant person can drink small amounts of milk and not show symptoms. (2)

- They have small amounts of lactase enzyme
- Which can effectively breakdown the small amount of milk the person ingested

2.2.2 Explain what you think will happen to the rate of the reaction if we increase the amount of milk the lactose intolerance patient drinks (3)

- The rate of reaction will increase
- Until all the enzymes are used
- Then the person will not be able to breakdown the excess lactose and will show symptoms of lactose intolerance

2.3 Draw a diagram showing the enzyme lactase and how it reacts with its substrate. Show the whole reaction process. (9)

Heading: 2 marks
Correct diagram showing the whole rxn: 3 marks
Label products (galactose and glucose): 1 mark
Label reactants (lactose and lactase): 1 mark
Neatness: 1 mark
Drawing rules= 1
[8] marks total

2.4 In your diagram, what is the name of the substrate? (1)

- Lactose

2.5 According to the article which population group is least likely to have lactose intolerance? (1)

- Caucasians

2.6 Draw a graph showing the percentages of different people groups inheriting lactose intolerance.

Heading= 2
Label X- axis= 1
Label Y-axis + (%)= 2
Graph= 4 (-1 if 1 graph is not correct)
Neatness= 1
Only 20% of Caucasians inherit lactose intolerance.

(10)

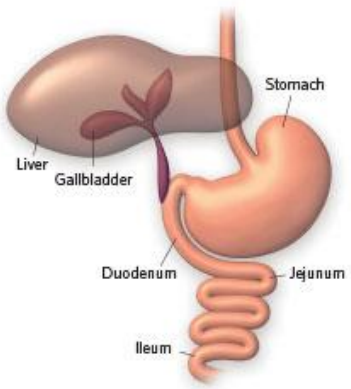
[28]

Question 3

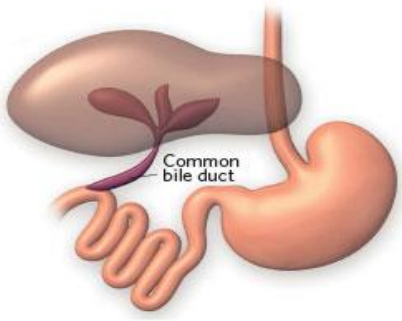
Read the text below and answer the following questions

Bile Benefits, Ruth Williams
Diverting the bile duct around a long stretch of the small intestine could treat obesity without cutting out chunks of the digestive tract.

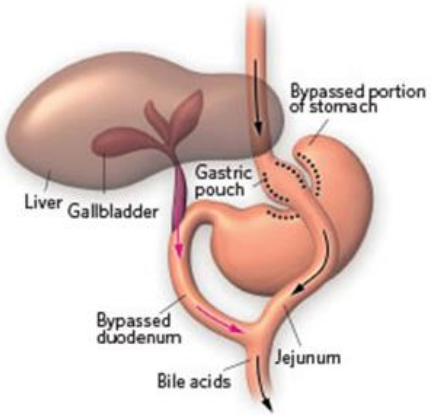
For obese patients undergoing a gastric bypass, such as the commonly used Roux-en-Y procedure, the surgery is extensive. “It basically changes the whole anatomy of the GI tract,” says Naji of Vanderbilt University. To scale down such procedures, Naji and his colleagues developed a new surgery in which the bile duct, which normally empties into the duodenum, is diverted farther down the small intestine to the ileum. In obese mice, this leads to weight loss, lower cholesterol, reduced blood sugar, increased energy expenditure and in some patients, reverses type 2 diabetes independent of weight loss- effects similar to those achieved with gastric bypass—without chopping up the intestine.



THE BEFORE: This diagram represents a presurgical mouse digestive system



BILE DIVERSION: To encourage bile acid absorption and recirculation, the common bile duct is disconnected from the duodenum and reconnected at the ileum. Metabolic and weight-loss benefits are similar to those of mice who receive the traditional Roux-en-Y procedure



ROUX-EN-Y: This common procedure to treat obesity shrinks the stomach and redirects the intestine to reduce calorie absorption. The duodenum, carrying digestive juices from the bile duct, is connected to the jejunum below where it has been connected to the gastric pouch.

https://www.the-scientist.com/bile-benefits-34584

3.1 Other than producing bile, what are two other functions of the liver? (2)

- Detoxifies/removes Drugs and Alcohol
- Stores Glycogen
- Activates vitamin D
- Metabolizes absorbed food molecules
- Any two

3.2 Bile plays a crucial role in the breaking down of fats. Explain the difference between saturated fats and unsaturated fats. (4)

- Saturated fats:
  - No double bonds
  - Solid at room temperature
  - Usually animal fats
  - Unhealthy fats
- Unsaturated fats:
  - Double bonds
  - Liquid at room temperature
  - Usually plant fats
  - Healthier fats

3.3 Benefits of the bile diversion surgery in some mice included the reversal of type 2 diabetes. Describe type 2 diabetes. (2)

- Body is unable to respond to insulin [insulin insensitivity or insulin resistance]
- blood glucose levels rise

3.4 In point form, explain how the body regulates blood glucose levels using two hormones. (8)

- One mark for each point:
- High blood glucose
- Pancreas secretes insulin
- Insulin converts Glucose to glycogen
- Which is stored in the liver/ muscles
- Decreases blood glucose levels
- Low blood glucose
- Pancreas secretes glucagon
- Glucagon converts glycogen in liver to glucose
- Thus increasing blood glucose levels
-

3.5 The Roux-en-Y procedure to treat obesity shrinks the stomach and redirects the intestine to reduce calorie absorption. Explain how the shrinking of the stomach reduces calorie absorption. (3)

_____

- Less food goes to the stomach
- Less food is broken down
- Less surface area for absorption
-

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Question 4

4.1 Read the article below and answer the questions that follow.

Climate Change May Favor Nitrogen-Fixing Plants
Aridity appears to configure landscapes with a greater diversity of plant species that rely on symbiotic bacteria for nitrogen.
Andy Carstens
Oct 17, 2022

In Death Valley National Park, mesquite plants (genus Prosopis) thrive in extreme aridity (very dry). While most vegetation types must extract most of their nutrients from fertile soil, mesquites and similar plants receive additional nitrogen from symbiotic bacteria, which enzymatically fix atmospheric nitrogen into an easily absorbed form in exchange for sugars produced during photosynthesis.



Because these plants have access to atmospheric nitrogen from their symbiotic bacteria, their leaves contain more nitrogen than other plants, and this buffers them against aridity by helping them retain water, says Mark Adams, an ecologist at the Swinburne University of Technology in Australia. When plants open their stomata to take in carbon dioxide, water escapes, but nitrogen stimulates the production of enzymes that improve the efficiency of carbon uptake, shortening how long plants need to hold their stomata open, Adams explains. “And that’s the secret [of] nitrogen-fixing plants.”

These findings add to prior evidence suggesting that nitrogen-fixing plants will outcompete non-fixers as the planet continues to warm, says Adams. However, future research should assess whether fixers generate a nitrogen surplus that could fertilize the soil, helping other plants withstand harsher conditions too, he adds. “That’s [the] bigger question.”

https://www.the-scientist.com/climate-change-may-favor-nitrogen-fixing-plants-70516

4.1.1 Mention one adaptation that helps the mesquite plants in the Death Valley National Park thrive in very dry conditions. (2)

- Their ability to fix nitrogen through symbiotic bacteria
- This adaptation allows them to access atmospheric nitrogen

4.1.2 Plants have two specialized tissues that play an important role in photosynthesis, the palisade mesophyll and spongy mesophyll. Give two adaptations for each tissue that enables them to effectively perform their functions. (4)

- Palisade mesophyll
  - Cells are elongated and packed tightly in rows to increase surface area
  - Each palisade mesophyll cell contains many chloroplasts
- Spongy mesophyll
  - contains many chloroplasts
  - Loosely packed irregular shaped cells allow for rapid diffusion of carbon dioxide to all photosynthesizing cells

4.1.3 How do the “non-fixers” acquire nitrogen from the atmosphere if they lack nitrogen-fixing bacteria? (1)

- When there’s lightning, the lightning fixes the atmospheric nitrogen for them
- They use the nitrogen that has been fixed by the nitrogen fixing bacteria found in other plants

4.1.4 What is the name of the symbiotic relationship between the mesquite plant and the nitrogen-fixing bacteria? (1)

- Mutualism

4.1.5 Discuss how an increase in nitrogen-fixing plants in the Death Valley National Park could influence the ecosystem. (4)

- Enhanced photosynthesis
- Which means more carbon dioxide would be absorbed and more oxygen released into the atmosphere
- These plants can potentially change the soils composition by increasing the nitrogen in the soil
- Thus benefiting other plants
- Any other logical

4.1.6 To which trophic level does the mesquite plant belong to? (1)

- Level 1/ Producers

4.1.7 State whether the following statement is true or false.

4.1.7.1 The abiotic factors in this ecosystem include the nitrogen fixing bacteria. (1)

- False

4.2 The xylem and phloem are responsible for transportation in plants, especially plants in dry areas. Tabulate 2 differences a xylem tissue and a phloem tissue. (5)

Table= 1
Heading= 1
4 points (2 each)= 4

• Xylem • Made up of dead cells • Transports water and minerals • Does not require ATP for transport • Phloem • Made up of living cells • Transports food

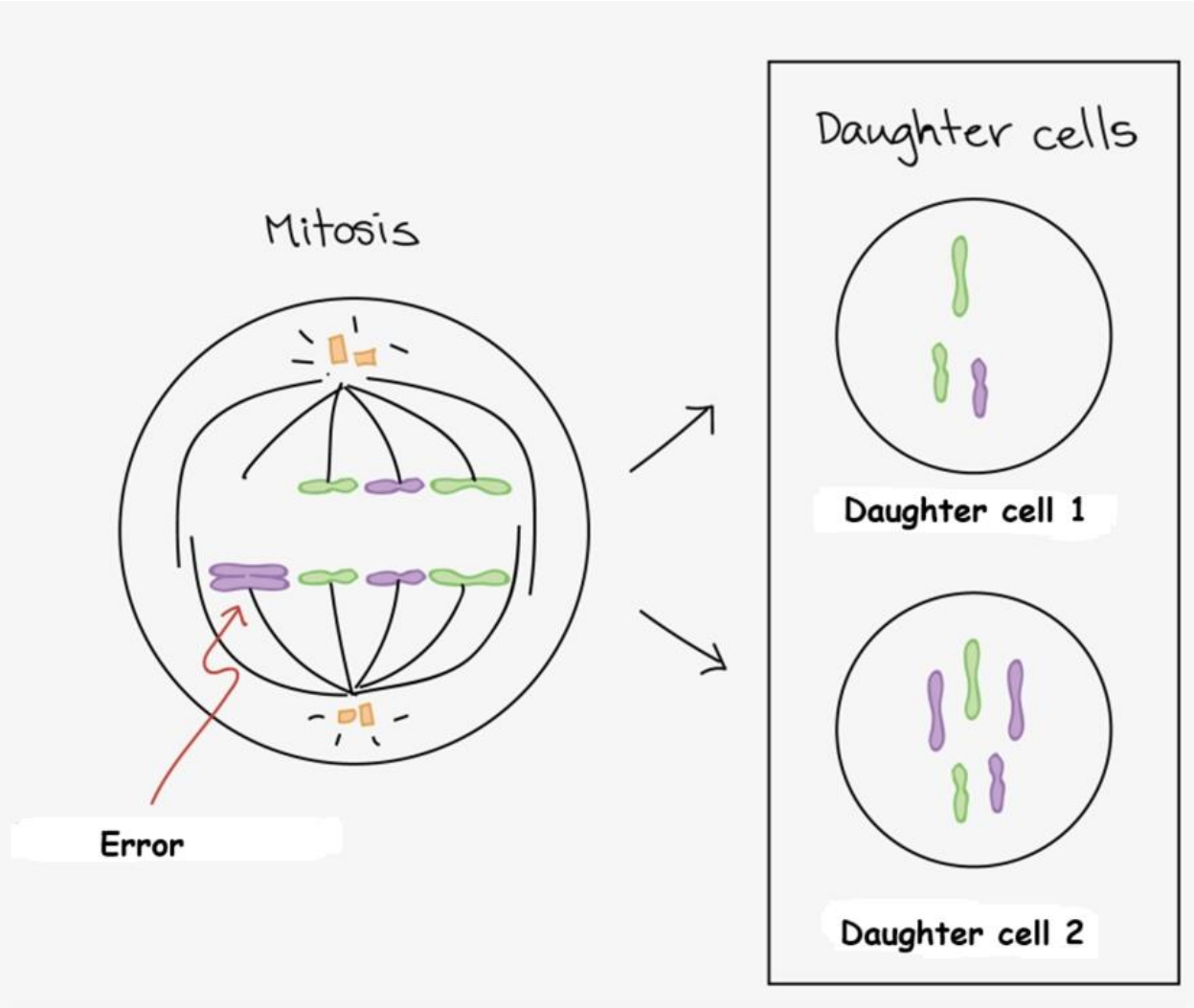
4.3 The opening and closing of stomata depends on many factors. Explain how osmosis controls the opening of the stomata. (4)

- Hypertonic condition in guard cell
- Water diffuses into cell
- Guard cell swells
- Opening the stomata

[24]

Question 5

The picture below shows an error in mitosis which results in an unequal distribution of chromosomes in daughter cells. Study the picture below and answer the following questions.



5.1 In what phase of mitosis did the error occur? (1)

_____

- Anaphase

5.2 In a normal healthy cell, what happens during the phase you mentioned in 5.1? (3)

_____

- Spindle fibres shorten
- Splitting the centromere and separating the chromatids
- Chromatids move to opposite poles

5.4 Name and explain what happens in the phase that follows the one you mentioned in 5.2. (5)

- Telophase:
- Spindle fibres are broken down
- Two new nuclei form
- Nuclear membranes and nucleoli reappear
- Cytokinesis divides the cell, we end up with two new cells

5..5 Draw the parent cell in Prophase of mitosis. (5)

Heading= 1 Correct drawing= 2 Correct number of chromosomes= 1 Neatness= 1

[14]